CECS 326 – Homework 3
CPU Scheduling Test Cases

These test cases are designed for the provided CPU scheduling simulator. Run each test using the default Round Robin time quantum of 2:

```
./scheduler input_file.txt
```

Test Case 1: All Processes Arrive at Time 0

Aspect tested: This case tests the basic ordering behavior of all four algorithms when every process is immediately available. FCFS should preserve input order, SJF should select the shortest burst, Priority should select the smallest priority number, and Round Robin should rotate through all processes.

Input

```
P1 0 5 3
P2 0 2 1
P3 0 4 2
```

Complete Expected Output

```
========================================
FCFS
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Gantt Chart:
| P1 | P2 | P3 |
0    5    7   11

Process   Arrival   Burst     Priority  Waiting     Turnaround    Response
---------------------------------------------------------------------------
P1        0         5         3         0           5             0
P2        0         2         1         5           7             5
P3        0         4         2         7           11            7

Average Waiting Time: 4.00
Average Turnaround Time: 7.67
Average Response Time: 4.00
```

```
CPU Utilization: 100.00%

========================================
SJF (Non-preemptive)
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Gantt Chart:
| P2 | P3 | P1 |
0    2    6   11

Process   Arrival   Burst     Priority  Waiting     Turnaround    Response
---------------------------------------------------------------------------
P1        0         5         3         6           11            6
P2        0         2         1         0           2             0
P3        0         4         2         2           6             2

Average Waiting Time: 2.67
Average Turnaround Time: 6.33
Average Response Time: 2.67
CPU Utilization: 100.00%

========================================
Priority (Non-preemptive)
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Gantt Chart:
| P2 | P3 | P1 |
0    2    6   11

Process   Arrival   Burst     Priority  Waiting     Turnaround    Response
---------------------------------------------------------------------------
P1        0         5         3         6           11            6
P2        0         2         1         0           2             0
P3        0         4         2         2           6             2

Average Waiting Time: 2.67
Average Turnaround Time: 6.33
Average Response Time: 2.67
CPU Utilization: 100.00%

========================================
Round Robin
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Gantt Chart:
| P1 | P2 | P3 | P1 | P3 | P1 |
0    2    4    6    8   10   11
```

```
Process   Arrival   Burst     Priority  Waiting     Turnaround    Response
-------------------------------------------------------------------------
P1        0         5         3         6           11            0
P2        0         2         1         2           4             2
P3        0         4         2         6           10            4

Average Waiting Time: 4.67
Average Turnaround Time: 8.33
Average Response Time: 2.00
CPU Utilization: 100.00%
```

Test Case 2: Staggered Arrival Times

Aspect tested: This case checks whether the simulator considers only processes that have already arrived. It also verifies that non-preemptive SJF and Priority do not interrupt a running process when a shorter or higher-priority process arrives later.

Input

```
P1 0 6 3
P2 1 2 1
P3 2 4 2
P4 4 1 1
```

Complete Expected Output

```
========================================
FCFS
========================================

Gantt Chart:
| P1 | P2 | P3 | P4 |
0    6    8   12   13

Process   Arrival   Burst     Priority  Waiting     Turnaround    Response
-------------------------------------------------------------------------
P1        0         6         3         0           6             0
P2        1         2         1         5           7             5
P3        2         4         2         6           10            6
P4        4         1         1         8           9             8

Average Waiting Time: 4.75
Average Turnaround Time: 8.00
Average Response Time: 4.75
CPU Utilization: 100.00%

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```

```
SJF (Non-preemptive)
========================================

Gantt Chart:
| P1 | P4 | P2 | P3 |
0    6    7    9   13

Process   Arrival   Burst     Priority  Waiting     Turnaround    Response
--------------------------------------------------------------------------
P1        0         6         3         0           6             0
P2        1         2         1         6           8             6
P3        2         4         2         7           11            7
P4        4         1         1         2           3             2

Average Waiting Time: 3.75
Average Turnaround Time: 7.00
Average Response Time: 3.75
CPU Utilization: 100.00%

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Priority (Non-preemptive)
========================================

Gantt Chart:
| P1 | P2 | P4 | P3 |
0    6    8    9   13

Process   Arrival   Burst     Priority  Waiting     Turnaround    Response
--------------------------------------------------------------------------
P1        0         6         3         0           6             0
P2        1         2         1         5           7             5
P3        2         4         2         7           11            7
P4        4         1         1         4           5             4

Average Waiting Time: 4.00
Average Turnaround Time: 7.25
Average Response Time: 4.00
CPU Utilization: 100.00%

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Round Robin
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Gantt Chart:
| P1 | P2 | P3 | P1 | P4 | P3 | P1 |
0    2    4    6    8    9   11   13

Process   Arrival   Burst     Priority  Waiting     Turnaround    Response
```

```
------------------------------------------------------------------------
P1        0         6         3         7           13          0
P2        1         2         1         1           3           1
P3        2         4         2         5           9           2
P4        4         1         1         4           5           4

Average Waiting Time: 4.25
Average Turnaround Time: 7.50
Average Response Time: 1.75
CPU Utilization: 100.00%
```

Test Case 3: CPU Idle Periods

Aspect tested: This case verifies that the simulator inserts IDLE intervals when no process is ready. It tests an initial idle period and another idle gap between process groups. CPU utilization should therefore be less than 100 percent.

Input

```
P1 2 3 2
P2 7 2 1
P3 7 4 3
```

Complete Expected Output

```
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FCFS
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Gantt Chart:
| IDLE | P1 | IDLE | P2 | P3 |
0      2    5      7    9   13

Process   Arrival   Burst     Priority  Waiting     Turnaround    Response
------------------------------------------------------------------------
P1        2         3         2         0           3             0
P2        7         2         1         0           2             0
P3        7         4         3         2           6             2

Average Waiting Time: 0.67
Average Turnaround Time: 3.67
Average Response Time: 0.67
CPU Utilization: 69.23%

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SJF (Non-preemptive)
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```

```
Gantt Chart:
| IDLE | P1 | IDLE | P2 | P3 |
0      2    5      7    9   13

Process   Arrival   Burst     Priority  Waiting     Turnaround    Response
---------------------------------------------------------------------------
P1        2         3         2         0           3             0
P2        7         2         1         0           2             0
P3        7         4         3         2           6             2

Average Waiting Time: 0.67
Average Turnaround Time: 3.67
Average Response Time: 0.67
CPU Utilization: 69.23%

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Priority (Non-preemptive)
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Gantt Chart:
| IDLE | P1 | IDLE | P2 | P3 |
0      2    5      7    9   13

Process   Arrival   Burst     Priority  Waiting     Turnaround    Response
---------------------------------------------------------------------------
P1        2         3         2         0           3             0
P2        7         2         1         0           2             0
P3        7         4         3         2           6             2

Average Waiting Time: 0.67
Average Turnaround Time: 3.67
Average Response Time: 0.67
CPU Utilization: 69.23%

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Round Robin
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Gantt Chart:
| IDLE | P1 | IDLE | P2 | P3 |
0      2    5      7    9   13

Process   Arrival   Burst     Priority  Waiting     Turnaround    Response
---------------------------------------------------------------------------
P1        2         3         2         0           3             0
P2        7         2         1         0           2             0
P3        7         4         3         2           6             2
```

```
Average Waiting Time: 0.67
Average Turnaround Time: 3.67
Average Response Time: 0.67
CPU Utilization: 69.23%
```

Test Case 4: Tie-Breaking

Aspect tested: This case checks deterministic tie-breaking. Processes with equal burst times, priorities, and arrival times should be selected according to their original input order. It also tests how Round Robin adds processes that arrive while another process is running.

Input

```
P1 0 4 2
P2 0 4 2
P3 1 2 1
P4 1 2 1
```

Complete Expected Output

```
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FCFS
========================================

Gantt Chart:
| P1 | P2 | P3 | P4 |
0    4    8   10   12

Process   Arrival   Burst     Priority  Waiting     Turnaround    Response
----------------------------------------------------------------------------
P1        0         4         2         0           4             0
P2        0         4         2         4           8             4
P3        1         2         1         7           9             7
P4        1         2         1         9           11            9

Average Waiting Time: 5.00
Average Turnaround Time: 8.00
Average Response Time: 5.00
CPU Utilization: 100.00%

========================================
SJF (Non-preemptive)
========================================

Gantt Chart:
| P1 | P3 | P4 | P2 |
```

```
0    4    6    8   12

Process   Arrival   Burst     Priority  Waiting     Turnaround    Response
--------------------------------------------------------------------------
P1        0         4         2         0           4             0
P2        0         4         2         8           12            8
P3        1         2         1         3           5             3
P4        1         2         1         5           7             5

Average Waiting Time: 4.00
Average Turnaround Time: 7.00
Average Response Time: 4.00
CPU Utilization: 100.00%

========================================
Priority (Non-preemptive)
========================================

Gantt Chart:
| P1 | P3 | P4 | P2 |
0    4    6    8   12

Process   Arrival   Burst     Priority  Waiting     Turnaround    Response
--------------------------------------------------------------------------
P1        0         4         2         0           4             0
P2        0         4         2         8           12            8
P3        1         2         1         3           5             3
P4        1         2         1         5           7             5

Average Waiting Time: 4.00
Average Turnaround Time: 7.00
Average Response Time: 4.00
CPU Utilization: 100.00%

========================================
Round Robin
========================================

Gantt Chart:
| P1 | P2 | P3 | P4 | P1 | P2 |
0    2    4    6    8   10   12

Process   Arrival   Burst     Priority  Waiting     Turnaround    Response
--------------------------------------------------------------------------
P1        0         4         2         6           10            0
P2        0         4         2         8           12            2
P3        1         2         1         3           5             3
P4        1         2         1         5           7             5
```

```
Average Waiting Time: 5.50
Average Turnaround Time: 8.50
Average Response Time: 2.50
CPU Utilization: 100.00%
```